

arrays. NASA along with its partners collaborated on a way to mount the solar arrays in a “blanket” which can be folded like an accordion and then deployed once it’s in space. Once it is in orbit, commands are sent to deploy the “blankets” to their full size. Gimbals are used to rotate these arrays so they can face the Sun to supply the maximum electrical power to the spacecraft.

With all this said, solar power doesn’t necessarily work the same for all spacecraft. As a spacecraft travels farther from the Sun, the solar power availability reduces and can become less efficient. On planetary missions, these spacecraft are affected by a planet’s weather conditions which will affect the solar panels, making it harder to obtain the light and radiation from the Sun which is required to convert solar energy to electrical power.

In cases where solar power doesn’t work as expected, scientists have developed different ways to power a spacecraft. One such way is the use of batteries which can store the power and save it for later.

BATTERY-POWERED SPACECRAFT

In the case of missions which are expected to be completed in a short period of time, the spacecraft doesn’t require a solar-powered source. In these situations, batteries need to provide enough energy for the spacecraft.

Spacecraft batteries are designed in a way to withstand tough environmental conditions as they will have to work on the surface of other worlds. These batteries are also required to be recharged many times. NASA scientists have developed ways in which batteries remain sustainable in such conditions and also store more energy and power in smaller sizes.

Nickel-hydrogen batteries provide a type of energy storage system where they can be recharged over 50,000 times over a 15-year cycle. These batteries

aren’t like the usual commercial batteries that we are used to, and also unlike commercial batteries which do not work in space, these are sealed systems which work exceptionally well in a vacuum.

NASA has also been working on new types of batteries in lieu of its impending Europa Mission, to one of Jupiter’s moons. These batteries are equipped to operate between -80°C (-112°F) and -100°C (-148°F). Advanced lithium-ion batteries with double the energy storage capacity are also being developed. This increased energy density, for example, would allow astronauts to spend twice as much time on the moon before the battery ran out.

ATOM-POWERED SPACECRAFT

Atoms make up anything and everything that exists in the universe. Atoms require and store

Saturn, Pluto and in its travel to interstellar space.

Radioisotope systems can function for a very long time like the two Voyager spacecrafts for example, which are still sending back information after 40+ years in space. These are solid-state devices, which means they don’t have any moving parts. They produce heat as a result of the radioactive decay of elements such as plutonium and have a quintessential lifespan of more than 30 years.

A Radioisotope Heater Unit employs a small pellet of plutonium dioxide to generate heat for the spacecraft structures, instruments and systems, enabling successful functioning throughout the period of the mission. For extra heat, some missions use only a few RHUs, while others use dozens. NASA has also investigated the possibility of using the same small



NASA's Curiosity rover uses both a lithium-ion battery and a radioisotope power system to function.

plenty of energy to hold themselves together. Albeit, there are some atoms which just aren’t stable enough to hold themselves together. These atoms are called radioisotopes, which fall apart much more easily. When these unstable atoms fall apart, they release energy in the form of heat.

A radioisotope powered system uses the heat that is generated from the unstable atom and the cold from space to generate electricity to power the spacecraft. This is a popular way of powering a spacecraft as NASA has previously used it in its missions to

fuel pellet used in an RHU to power a compact system capable of producing a few dozen milliwatts of electrical power.

Over the years, NASA has investigated other types of nuclear power technology, such as space nuclear reactors and nuclear propulsion technologies. Continued research and development of these and other related technologies may one day allow space missions to deliver more inventory on cargo missions, obtain faster trip times on piloted missions, or even power crew stations on Mars or the Moon’s surface. **d**